

Real-time PwD Scenarios + Phase 1 / Phase 2 Work

How NeuroShift works for different motor disabilities, and what major work happens in each semester | July 2026

NeuroShift helps people who still have some voluntary muscle signal. It is a home-control assistant (Act/Abstain), not a cure for paralysis and not full-body movement restoration.

1. Will there be major work in BOTH phases?

YES. Both semesters have major, different work. Neither is 'light'.

	Phase 1 (Sem 7)	Phase 2 (Sem 8)
Nature	BUILD the system	PROVE the research
Major work	Hardware + software pipeline + live demo	Fusion/abstention science + evaluation + paper
Hard parts	EMG noise, wiring, YOLO desk scene, MQTT reliability	FAR tuning, priors, ablations, writing IEEE draft
Success looks like	External sees lamp/fan Act and Abstain live	Tables show lower FAR vs baselines + paper draft
Effort level	High (engineering heavy)	High (research + polish heavy)

Phase 1 major work checklist

- MyoWare + ESP32 streaming and calibration
- Relay/smart-plug control for normal Indian wired appliances
- YOLO/RT-DETR + MediaPipe affordance pipeline
- Rule-based intention policy (Act/Abstain)
- Dashboard + synchronized trial logger
- Pilot dataset start + 3 baselines
- Rehearsed live external demo

Phase 2 major work checklist

- Learned multimodal fusion (replace pure rules)
- Calibrated abstention tuned for False Actuation Rate (FAR)
- Personal temporal prior (time/habit)
- Decision-level explanations + caregiver dashboard
- Full evaluation + ablations on NeuroShift-ADL
- IEEE paper draft + final report/viva

Phase 1 ownership: We built a working EMG+vision+IoT intention prototype with Act/Abstain. | Phase 2 ownership: We proved lower FAR with fusion + abstention + prior, and wrote the paper.

2. Common real-time loop (all users)

1. Camera sees room + devices + body/hand pose (if visible)
2. System lists feasible actions (lamp / fan / plug)
3. EMG checks remaining voluntary muscle effort
4. Policy decides: ACT one device OR ABSTAIN
5. ESP32 / MQTT controls that appliance
6. Dashboard logs why

3. Real-time scenarios by PwD type

3.1 Mild hand weakness / tremor / arthritis

EMG: forearm or biceps. Vision: medium importance.

- User orients to lamp and slightly raises arm.
- Camera marks lamp as feasible; EMG detects effort.
- Decision: ACT -> lamp ON.
- If lamp+fan both unclear: ABSTAIN.

Fit: Excellent (best Phase 1 target).

3.2 Hemiparesis (one side weak after stroke)

EMG on stronger side residual muscle (forearm/biceps/shoulder).

- User turns toward fan; tenses good-side muscle.
- Camera supports fan candidate; EMG confirms volition.
- Decision: ACT -> fan ON.
- Accidental tensing without clear target -> ABSTAIN.

Fit: Very good.

3.3 Upper-limb amputation

EMG on residual stump. Closest to base-paper users, but output is home devices.

- User faces bedside lamp.
- Camera: lamp candidate. User contracts residual muscles.
- Decision: ACT -> lamp ON.
- No need for gesture dictionary (gesture1/gesture2).

Fit: Excellent; strong extension of Chamberland EMBC 2024.

3.4 Partial upper-body paralysis

EMG on best residual site (trapezius/deltoid/biceps). Vision does more work.

- Camera views bed + lamp + fan.
- User orients to lamp and gives small shoulder shrug.
- If one device clearly favored: ACT. Else ABSTAIN/clarify.

Fit: Good, if a stable EMG site exists.

3.5 Severe quadriplegia / shoulder-to-leg paralysis

Honest limit: arm MyoWare may fail if no usable limb EMG. Works only if residual neck/shoulder/face EMG exists. Otherwise need future modalities (gaze/voice/sip-puff) - not Phase 1 claim.

- If residual EMG exists and vision strongly favors one device: ACT.
- Otherwise: ABSTAIN (critical for safety).

Fit: Conditional. State this clearly to guide/external.

3.6 Paraplegia (legs paralyzed, arms OK)

Arms often usable; issue is reach/wheelchair access/fatigue.

- User cannot reach wall switch from wheelchair.
- Orients to fan + intentional forearm contraction.
- Decision: ACT -> fan.

Fit: Good for independence on unreachable switches.

3.7 Cerebral palsy / inconsistent motor control

Involuntary spikes can create false EMG - Act/Abstain is essential.

- Involuntary jerk + no clear visual target -> ABSTAIN.
- Later intentional orientation to lamp + clearer pattern -> ACT.

Fit: Medium-good with strict FAR tuning.

3.8 Elderly with low vision + weak hands

- User wants night lamp; cannot use remote buttons easily.
- Camera finds bedside lamp; EMG confirms effort -> ACT.
- Caregiver dashboard shows decision log.

Fit: Good.

4. Quick fit matrix

Condition	EMG placement	Vision role	Works now?
Hand weakness / tremor	Forearm	Medium	Yes
Stroke hemiparesis	Stronger limb	High	Yes
Arm amputee	Residual stump	High	Yes
Partial upper paralysis	Shoulder/upper arm	Very high	Yes if signal
Paraplegia (arms OK)	Forearm	Medium	Yes
Severe quadriplegia	Neck/face if available	Very high	Only if EMG exists
No voluntary EMG at all	-	-	No (need other modalities)

5. Day-in-life example (stroke, weak right arm)

- Evening: lamp and fan both visible.
- User turns to lamp and contracts left forearm.
- Fusion: lamp high, fan low, EMG high -> ACT lamp.
- Later: shifts in chair (EMG noise), no clear target -> ABSTAIN.
- Dashboard: 'Abstain reason: ambiguity + low confidence'.

This is novelty in daily life: intention when clear, safety when unclear.

6. What to claim vs not claim

Correct claim

NeuroShift is for people with motor impairment who retain some voluntary muscle activity, enabling safer smart-home control through scene understanding + EMG confirmation + abstention.

Incorrect claim

It works for all paralysis types automatically / restores full movement.

7. Which users for which phase

Phase	Practical users
Phase 1 demo	Able-bodied pilot + simulated weakness; stump/forearm EMG
Phase 2 study	Consenting volunteers: hemiparesis / amputation / elderly weak grip
Future work	Gaze/voice fallback for severe quadriplegia without limb EMG

8. Bottom line

- YES - major work in Phase 1 AND Phase 2 (different kinds of major).
- Phase 1 = build reliable real-time demo for Indian wired homes.
- Phase 2 = prove FAR reduction and write IEEE paper.
- Best-fit users: residual voluntary EMG + need for lamp/fan/plug control.
- Severe complete paralysis with zero EMG needs extra modalities later.

Real-time promise: when intention is clear, Act. When unsure, Abstain. Measure success mainly by False Actuation Rate.